ISE 3200 - HW1: Modeling & the Graphical Method

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Q1. Buckeye Furniture makes tables and chairs. A table needs 4 hours of carpentry and 2 hours of finishing; a chair needs 3 hours of carpentry and 1 hour of finishing. Each week 240 carpentry hours and 100 finishing hours are available. Each table earns $70 profit and each chair $50. Formulate an LP that maximizes weekly profit.

Q1a) Define the decision variables (with units).

x1 = tables built per week, x2 = chairs built per week

Q1b) Write the objective function.

max z = 70x1 + 50x2

Q1c) Write all constraints.

4x1 + 3x2 <= 240 (carpentry); 2x1 + x2 <= 100 (finishing); x1, x2 >= 0

Q2. Solve the LP from Q1 graphically.

Q2a) List every corner point of the feasible region.

(0, 0), (50, 0), (30, 40), (0, 80)

Q2b) Give the optimal solution and its profit.

x1 = 30, x2 = 40, z = $4,100

Q3) Suppose the finishing constraint is removed. Is the LP still bounded? Give the new optimum.

Still bounded: carpentry limits both products. New optimum x1 = 60, x2 = 0, z = $4,200.